WORKED SOLUTION

STEP III 2004, Question 7

Pure • Reduction Formulae

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q7

The key idea

The integrand of I_{n+1} is the integrand of I_n multiplied by $\frac{t}{t+1}$, so a bound on that one factor gives a bound linking consecutive integrals. Integration by parts gives an exact link too, and putting the inequality and the equality side by side squeezes out a bound on I_n alone. The exact link telescopes down to I_1, which is a logarithm we know, and that turns everything into a series for $\ln 2$ with an error term we can control.

Solution

The inequality $I_{n+1} < \frac{1}{2} I_n$

On $0 \leq t \leq 1$,

$$\frac{t}{t+1} = 1 - \frac{1}{t+1},$$

which increases with t, so its greatest value is $\frac{1}{2}$, taken only at $t = 1$. Now write the integrand of I_{n+1} as

$$\frac{t^n}{(t+1)^{n+1}} = \frac{t}{t+1} \cdot \frac{t^{n-1}}{(t+1)^n}.$$

Hence

$$\frac{1}{2} I_n - I_{n+1} = \int_0^1 \left(\frac{1}{2} - \frac{t}{t+1}\right) \frac{t^{n-1}}{(t+1)^n} \, \mathrm{d}t.$$

The integrand here is continuous, never negative, and strictly positive for $0 < t < 1$ (both factors are positive there). So the integral is strictly positive, and $I_{n+1} < \frac{1}{2} I_n$.

? **Ask yourself** Why is the inequality strict, when $\frac{t}{t+1}$ actually reaches $\frac{1}{2}$?

It reaches $\frac{1}{2}$ only at the single point $t = 1$. A continuous non-negative function that is positive on a whole interval has a positive integral, so one point of equality makes no difference.

The recurrence

Integrate I_{n+1} by parts with $u = t^n$ and $\dfrac{\mathrm{d}v}{\mathrm{d}t} = (t+1)^{-(n+1)}$, so that $v = -\dfrac{1}{n(t+1)^n}$ (valid as $n \geq 1$):

$$\begin{aligned} I_{n+1} &= \Big[-\frac{t^n}{n(t+1)^n}\Big]_0^1 + \int_0^1 \frac{nt^{n-1}}{n(t+1)^n}\,\mathrm{d}t \\ &= -\frac{1}{n\,2^n} + I_n, \end{aligned}$$

since the bracket is $-\frac{1}{n2^n}$ at $t = 1$ and 0 at $t = 0$. This is the required result.

The bound on I_n

Substitute the recurrence into the inequality:

$$I_n - \frac{1}{n\,2^n} = I_{n+1} < \frac{1}{2}I_n \quad\Longrightarrow\quad \frac{1}{2}I_n < \frac{1}{n\,2^n} \quad\Longrightarrow\quad I_n < \frac{1}{n\,2^{n-1}}.$$

The series for $\ln 2$

First,

$$I_1 = \int_0^1 \frac{1}{t+1}\,\mathrm{d}t = [\ln(t+1)]_0^1 = \ln 2.$$

The recurrence says $I_r - I_{r+1} = \dfrac{1}{r\,2^r}$ for each $r \geq 1$. Adding these for $r = 1, \ldots, n$, the left-hand side telescopes:

$$I_1 - I_{n+1} = \sum_{r=1}^{n} \frac{1}{r\,2^r}, \qquad \text{so} \qquad \ln 2 = \sum_{r=1}^{n} \frac{1}{r\,2^r} + I_{n+1}.$$

The bounds $\frac{2}{3} < \ln 2 < \frac{17}{24}$

We have two facts about the remainder: $I_{n+1} > 0$, because its integrand is positive on $(0, 1]$, and $I_{n+1} < \dfrac{1}{(n+1)2^n}$, by the bound above with $n+1$ in place of n. So for every n,

$$\sum_{r=1}^{n} \frac{1}{r\,2^r} < \ln 2 < \sum_{r=1}^{n} \frac{1}{r\,2^r} + \frac{1}{(n+1)2^n}.$$

The partial sums are

$$\tfrac{1}{2}, \qquad \tfrac{1}{2} + \tfrac{1}{8} = \tfrac{5}{8}, \qquad \tfrac{5}{8} + \tfrac{1}{24} = \tfrac{16}{24} = \tfrac{2}{3}.$$

Ask yourself Which value of n should we use for each bound?

Different ones. For the lower bound take $n = 3$: $\ln 2 > \frac{2}{3}$. For the upper bound take $n = 2$, where the error bound is $\frac{1}{3\cdot 4} = \frac{1}{12}$:

$$\ln 2 < \tfrac{5}{8} + \tfrac{1}{12} = \tfrac{15}{24} + \tfrac{2}{24} = \tfrac{17}{24}.$$

Together, $\frac{2}{3} < \ln 2 < \frac{17}{24}$, as required.

Checks

Numerically $\ln 2 \approx 0.6931$, which does lie between $\frac{2}{3} \approx 0.6667$ and $\frac{17}{24} \approx 0.7083$. Simpson's rule gives $I_3 \approx 0.0681$, comfortably below the bound $\frac{1}{12} \approx 0.0833$, and indeed $\frac{5}{8} + 0.0681 = 0.6931$. We also checked the recurrence and both inequalities numerically for n up to 24.

Common slips

- Using the same n for both bounds: $n = 3$ gives the upper bound $\frac{2}{3} + \frac{1}{32} = \frac{67}{96}$, which is true and sharper, but it is not the $\frac{17}{24}$ asked for.
- Getting the sign of the boundary term wrong in the integration by parts; v is negative.
- Forgetting to say why $I_{n+1} > 0$, which the strict lower bound needs.

Beyond STEP

This is the series $\ln 2 = \sum_{r \geq 1} \frac{1}{r 2^r}$, which is $-\ln(1 - x) = \sum x^r / r$ at $x = \frac{1}{2}$, with an explicit bound on the remainder after n terms. Estimating the remainder by an integral is exactly how Taylor's theorem with integral remainder works, a staple of first-year analysis. See Abbott, *Understanding Analysis*, for the theory of power series and their remainders.

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